

TAE KOAN YOO

Seongnam, Republic of Korea

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RESEARCH INTERESTS

- Multimodal Representation Learning, Controllable Generative Models, Diffusion Models, Latent Space Geometry

EDUCATION

SUNGKYUNKWAN UNIVERSITY, Seoul, Republic of Korea

- **M.S.**, Electrical and Computer Engineering **02/2018-02/2020**
GPA: 4.04 / 4.5 (95.4%)
Advisor: Prof. Jong Tae Kim (College of Information and Communication Engineering)
Research Experience:
 - Conducted comprehensive research on deep learning optimization for embedded models, including pruning, quantization, and model compression.
 - Focused on quantization-based optimization and proposed Double-Stage Weight Sharing, reducing over 80% of model parameters while maintaining approximately 90% accuracy.
- **B.S.**, Electronic and Electrical Engineering **02/2011-02/2018**
GPA: 3.89 / 4.5 (93.9%), Major: 4.08 / 4.5, Upper-Division: 4.26 / 4.5 (97.6%)
Research Experience:
 - Conducted system-level research on the Linux EXT4-SSD FTL interface; proposed fTrim/vTrim schemes that reduced file-deletion latency by up to 63%, receiving the Best Undergraduate/Junior Paper Award.Additional Experience:
 - Exchange Student Program, The University of Texas at Austin, USA 01/2017-08/2017

WORK EXPERIENCE SUMMARY

- NHN CORPORATION**, Multi-modal Development Team, Seongnam, Republic of Korea **09/2023-Present**
- Research and development on generative models for artistic image, audio, and video creation.
- HYUNDAI MOTOR COMPANY**, Autonomous Driving SW Team, Seoul, Republic of Korea **01/2020-09/2023**
- Research and development on driving-scene understanding with vision-LiDAR perception.

PUBLICATIONS

- [1] **T. K. Yoo**, W. K. Jung, K. H. Kim, and K. B. Kong, "PADS-TAL: Padding Annealed Diffusion Sampling in Text-Aware Latent Space for Robust and Diverse Text-to-Music Generation," *International Conference on Machine Learning (ICML)* (2026)
- Led the project by formulating the core problem, proposing PADS and TAL, implementing the experimental pipeline, conducting the main experiments, and writing the manuscript.
- [2] **T. K. Yoo**, "Minimizing Required Data in a Neural Network Accelerator Using Double-Stage Weight Sharing," *Master's Thesis, Sungkyunkwan University* (2020)
- [3] **T. K. Yoo**, J. K. Park, and J. T. Kim, "VLSI Implementation of Area-Efficient Parallelized Neural Network Accelerator Using Hashing Trick," *International SoC Design Conference (ISODC)* (2019)
- [4] **T. K. Yoo**, G. H. Hong, and D. K. Shin, "Analysis on Accelerating Object Detection in Edge Device with Half-Precision Floating Point," *Korea Computer Congress (KCC)* (2018)
- [5] **T. K. Yoo**, K. H. Han, and D. K. Shin, "Optimizing Trim Operations by In-Storage Processing," *Korea Computer Congress (KCC)* (2017)

WORK EXPERIENCE

NHN CORPORATION, Seongnam, Republic of Korea

- **Senior Research Engineer**, Multi-modal Development Team, AI Tech Lab. **09/2023-Present**
Role: Generative AI research and service development for music and visual content creation
 - Led research and development of diffusion-based text-to-music generation models, covering data curation, model training, sampling strategies, and evaluation protocols.
 - Developed personalized text-to-image generation models for webtoon sub-character creation.Notable achievement:
 - Led the end-to-end development of a production-level generative AI service for music creation, resulting in three patent applications.
 - In independent research conducted during this period, proposed PADS-TAL for alignment-preserving diversity control in text-to-music generation; achieved a 15.4% gain in overall diversity and a 71.6% gain in within-condition diversity, leading to an *ICML* 2026 paper.External collaboration: NHN Bugs, NHN Comico
- **Senior Research Engineer**, Voice AI Team, AI Tech Lab. **11/2023-02/2024**
Role: Automatic Speech Recognition (ASR) model research and system optimization
 - Optimized ASR inference and streaming pipelines using TensorRT, CUDA memory management, IPC, multi-threading, and gRPC.
 - Reimplemented and optimized beam search for real-time speech recognition services.Notable achievement:
 - Reduced memory usage by 65% and processing time by 40% for streaming ASR deployment.

HYUNDAI MOTOR COMPANY, Seoul, Republic of Korea

- **Research Engineer**, Camera-LiDAR Fusion Group, Autonomous Driving SW Team **09/2021-09/2023**
Role: Camera-LiDAR fusion and multimodal perception for autonomous driving
 - Developed camera-LiDAR fusion networks for 3D object detection and road information recognition using internal autonomous-driving datasets.
 - Proposed a cross-attention-based fusion architecture to dynamically capture correlations between image and LiDAR modalities, improving mAP for long-range and small objects.
 - Built multimodal data pipelines for camera-LiDAR perception, covering sensor calibration, cross-sensor transformation, data collection, annotation management, and dataset quality control.Notable achievement:
 - Proposed the company's first fusion-based perception method, which was prepared for production-level deployment and resulted in two registered patents.External collaboration: NVIDIA Korea, AWS Korea
- **Research Engineer**, Road Information Detection Group, Autonomous Driving SW Team **01/2020-09/2021**
Role: Real-time computer vision and embedded inference for autonomous driving
 - Developed real-time computer vision pipelines and image-processing/post-processing algorithms for road information extraction from monocular camera images.
 - Optimized inference pipelines using TensorRT, CUDA, Nsight, TensorBoard, and multi-stream/asynchronous processing.Notable achievement:
 - Developed and optimized a road boundary recognition pipeline deployed in production vehicles.

PATENTS

- [1] **T. K. Yoo**, W. K. Jung, K. H. Kim, "Method and System for Generating Audio Source Incorporating User-Specified Melody Using a Diffusion Model," Korean Patent Application
- [2] K. H. Kim, **T. K. Yoo**, "Method and System for Generating Music Using Diffusion Model Based on BPM Measurement," Korean Patent Application
- [3] **T. K. Yoo**, K. H. Kim, "Method and System for Generating Music Using Diffusion Model Based on Genre-Style Structured Information," Korean Patent Application
- [4] J. H. Lim, H. C. Choi, J. H. Kim, and **T. K. Yoo**, "Object Recognition Apparatus and Vehicle Having the Same," US Patent Publication: US20240046658A1 (2024)
- [5] J. H. Kim, H. C. Choi, J. H. Lim, and **T. K. Yoo**, "Apparatus and Method for Object Recognition," US Patent Publication: US20240104935A1 (2024)

[6] J. T. Kim, **T. K. Yoo**, and D. Y. Kim, “Neural Network Accelerator Using Double-Stage Weight Sharing,” Korean Patent Publication: KR10-2519210 (2023)

HONORS AND AWARDS

- Excellent Technical Report Award, Hyundai Motor Company 2020
- Best Paper Award, Korea Computer Congress, KIISE 2018
- Best Undergraduate/Junior Paper Award, Korea Software Congress, KIISE 2018
- National Science and Engineering Scholarship, Korea Student Aid Foundation 2016
- Dean’s List, College of Information & Communication Engineering, Sungkyunkwan University 2016

ACTIVITIES

- *Undergraduate Research Student*, Embedded Software Lab., Sungkyunkwan University 08/2017-02/2018
- *Exchange Student*, The University of Texas at Austin, USA 01/2017-08/2017
- *Working Holiday Program*, Australia 08/2014-09/2015
- *Official University Ambassador*, Sungkyunkwan University, Republic of Korea 02/2011-11/2012

MILITARY SERVICE

- *Sergeant*, Republic of Korea Army, Republic of Korea 11/2012-08/2014